



THE
ZSMC CO
ESTD 2026



Appointment of Kumari Anushka

Document ID: ZSMC-REC-001 Rev 0

Effective Date: 17-08-2026

Evidence before prestige. Reproducibility before publicity. Intellectual contribution before title. Scientific honesty before institutional interest.

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Srijal Dutta
Founder
The ZSMC Co.



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Appointment

The ZSMC Co. hereby records the appointment of Kumari Anushka as a Researcher within The ZSMC Co. research program.

The appointment follows the review of her Research Program Application and subsequent applicant meeting. The application was assessed on the basis of the proposed research, scientific merit, feasibility, research integrity, and the applicant's ability to contribute to collaborative research activities.

The appointment is issued on a provisional basis for an initial period of two months, after which the appointment shall be reviewed.

The review period may be shortened or the appointment confirmed earlier where sufficient evidence of research contribution and suitability is available.

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Researcher Information

FULL NAME	Kumari Anushka
Role	Provisional Researcher
Appointment Status	Provisional (2 Months)
Email	kumari.anushka_ug2025@ashoka.edu.in
Country	India
ORCID	NA
Affiliation	The ZSMC Co.
Appointment Period	08 August 2026
Appointment Status	Meeting Completed

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Research Area

Kumari Anushka's research interests and proposed work primarily concern:

- Microbiology
- Microbial stress resilience
- Quantitative microbiology
- Microbial biotechnology
- Biological systems modelling
- Protein biology
- Computational biology
- Stress adaptation
- Experimental strain evaluation

Her principal research project at the time of appointment is:

The Stress Capacitance Index: Predicting Which Microbes Keep Working Under Fermentation Stress

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Principal Research Project

The Stress Capacitance Index (SCI) is an explainable screening framework intended to investigate whether microbial strains can be systematically evaluated according to their ability to retain functional performance under environmental and fermentation-related stress.

The project combines public biological and strain information with experimental measurements. The proposed framework considers characteristics associated with acid response, oxidative stress, protein folding, secretion, cellular repair, recovery, and reserve metabolism.

The research is intended to investigate whether such information can provide a defensible basis for selecting microbial strains before substantial optimization or genetic engineering is undertaken.

At the time of application, the project had progressed beyond the conceptual stage. Preliminary work included an experimental comparison of *Bacillus subtilis* and *Lactococcus lactis* under acidic stress and development of a public-data ranking system using resources including UniProt, NCBI, BacDive, and ATCC.

The next phase is intended to extend experimental validation to multiple authenticated *B. subtilis* strains and evaluate whether computational rankings correspond with measurable biological performance.



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Initial Appointment Basis

The appointment was supported by the following considerations:

1. The applicant demonstrated an established interest in early-career biological research.
2. The proposed research addresses a defined scientific and practical problem in microbial biotechnology.
3. The applicant had progressed beyond the idea stage and demonstrated preliminary experimental activity.
4. The applicant had developed an initial computational and public-data framework relevant to the proposed research.
5. The applicant demonstrated interest in combining wet-lab experimentation with quantitative and computational methods.
6. The proposed research is compatible with The ZSMC Co.'s interest in interdisciplinary and independently conducted scientific research.
7. The applicant acknowledged the organization's requirements concerning research integrity and accurate distinction between experimental, simulated, and theoretical results.

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Research Integrity

As part of the application, the applicant affirmed that:

- the information submitted in the application is truthful;
- acceptance is not guaranteed;
- research shall be conducted ethically and honestly;
- experimental results, simulations, and theoretical work shall be distinguished accurately; and
- The ZSMC Co. may decline or discontinue participation where these principles are not followed.

These commitments form part of the basis of the provisional appointment.

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Provisional Appointment Conditions

During the provisional appointment period, the researcher is expected to:

- conduct research in accordance with applicable ethical, safety, and institutional requirements;
- maintain accurate and reproducible research records;
- distinguish clearly between observations, experimental results, computational predictions, assumptions, and hypotheses;
- avoid presenting preliminary or unvalidated findings as established conclusions;
- appropriately document methods, datasets, analyses, and experimental conditions;
- communicate material research findings and limitations accurately;
- respect intellectual property, confidentiality, and authorship requirements applicable to The ZSMC Co. research activities;
- participate in reasonable research discussions, reviews, or reporting activities associated with the assigned work; and
- promptly disclose any material issue that may affect the integrity, safety, reproducibility, or interpretation of the research.

The provisional appointment does not constitute an employment relationship and does not provide a salary, stipend, guaranteed funding, or other financial compensation unless separately agreed in writing.

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Research Independence and Collaboration

The researcher may continue to pursue independent academic or research activities outside The ZSMC Co., subject to applicable intellectual-property, confidentiality, authorship, and conflict-of-interest requirements.

Participation in The ZSMC Co. does not by itself transfer ownership of independently developed work to the organization.

Where research is conducted collaboratively under The ZSMC Co., ownership, authorship, publication, data use, and intellectual-property arrangements should be established appropriately for the relevant project before material collaborative work is undertaken.

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Provisional Review

The provisional appointment shall ordinarily be reviewed after approximately **two months**.

The review may consider:

- quality and consistency of research work;
- scientific reasoning and methodological discipline;
- ability to maintain reliable research records;
- contribution to the assigned research project;
- responsiveness to scientific feedback;
- adherence to research-integrity requirements;
- ability to distinguish evidence from interpretation;
- collaboration and communication; and
- feasibility of continued participation within The ZSMC Co.

The review may result in one of the following outcomes:

A. Confirmed Researcher

The provisional appointment is successfully completed and the researcher continues under confirmed researcher status.

B. Extended Provisional Appointment

Additional time is required to assess research contribution or complete an appropriate research milestone.

C. Appointment Concluded

The provisional appointment is not continued.

The review outcome shall be recorded in an appropriate subsequent organizational record.

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Researcher Recognition

During the provisional period, Kumari Anushka may be identified publicly as:

Kumari Anushka — Provisional Researcher, The ZSMC Co.

Upon confirmation, the public designation may be updated to:

Kumari Anushka — Researcher, The ZSMC Co.

Public descriptions of the researcher's work shall accurately reflect the provisional or confirmed status of the appointment.

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Effective Date and Duration

Effective Date: 17 August 2026

Initial Provisional Period: Two months from the effective date, subject to the review provisions of this record.

The appointment may be reviewed earlier where appropriate and may be concluded earlier in accordance with organizational requirements or applicable agreements.

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Founder
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Record of Approval

This record constitutes the formal organizational record of the appointment of Kumari Anushka to the position of Researcher at The ZSMC Co.

Appointing Authority:

Srijal Dutta

Founder & Lead Researcher

The ZSMC Co.

Appointment Decision: ACCEPTED — PROVISIONAL

Date of Decision: 17 August 2026

Researcher Acknowledgement: Pending / To be recorded

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Srijal Dutta

Founder

The ZSMC Co.